

# Co-1 Assessment Tool-1

MLA0406

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- 1) A coin is tossed 10 times and the outcomes are

H H T H H H T H H T

using Maximum Likelihood Estimation,  
estimate the probability of getting  
Heads (P)

soln) outcomes  $\rightarrow H = 7$

$$T = 3$$

Total tosses = 10

$$\hat{P} = \frac{\text{Number of Heads}}{\text{Total tosses}}$$

$$\hat{P} = \frac{7}{10} = 0.7$$

$$P = 0.7 \text{ (70\%)}$$

- 2) out of 50 students, 42 passes an exam, Assuming a Bernoulli distribution, Find the MLE of probability)

sol)

Total no. of students ( $n$ ) = 50

no. of students who passed ( $x$ ) = 42

$$\hat{P} = \frac{x}{n}$$

$$\hat{P} = \frac{42}{50}$$

$$\hat{P} = 0.84$$

$$\hat{P} = 84\%$$

MLE Estimate of the probability of passing is 84%.

- 3) A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

sol)

Total no. of bulbs ( $n$ ) = 100

no. of defective bulbs ( $x$ ) = 8

$$\hat{P} = \frac{x}{n}$$

$$\hat{P} = \frac{8}{100}$$

$$\hat{p} = 0.08$$

$$\hat{p} = 8\%$$

MLE estimate of the probability of a bulb being defective is 8%.

- 4) A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Sol) Total no. of rolls ( $n$ ) = 60

no. of times 6 appears ( $x$ ) = 18

$$\hat{p} = \frac{x}{n}$$

$$\hat{p} = \frac{18}{60}$$

$$\hat{p} = 0.3$$

$$\hat{p} = 30\%$$

MLE estimate of the probability of rolling a 6 is 30%.

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$$\text{Input } (x) = 4$$

$$\text{Weight } (W) = 2$$

$$\text{Actual output} = 10$$

$$\text{Prediction: } y = W \times x$$

1. Find the Predicted output

2. Calculate the error

3. If the gradient is 4 and learning rate is 0.01, Find the updated weight.

sol) 1. Predicted output

$$y = W \times x$$

$$y = 2 \times 4$$

$$y = 8$$

$$\text{Predicted output} = 8$$

2. Calculate the error

$$\text{Error} = \text{Actual output} - \text{predicted output}$$

$$\text{Error} = 10 - 8$$

$$\text{Error} = 2$$

3. The updated weight

$$W_{\text{new}} = W - (\eta \times \text{gradient})$$

$$W = 2$$

$$\eta = 0.01$$

$$\text{Gradient} = 4$$

$$W_{\text{new}} = 2 - (0.01 \times 4)$$

$$W_{\text{new}} = 2 - (0.04)$$

$$W_{\text{new}} = 1.96$$

$$\text{updated weight} = 1.96$$